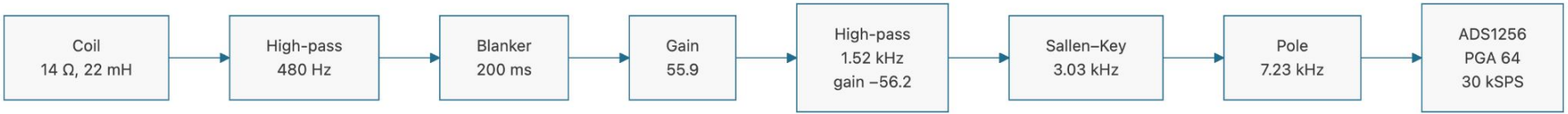
Signal Processing Pipeline

September 23rd, 2026

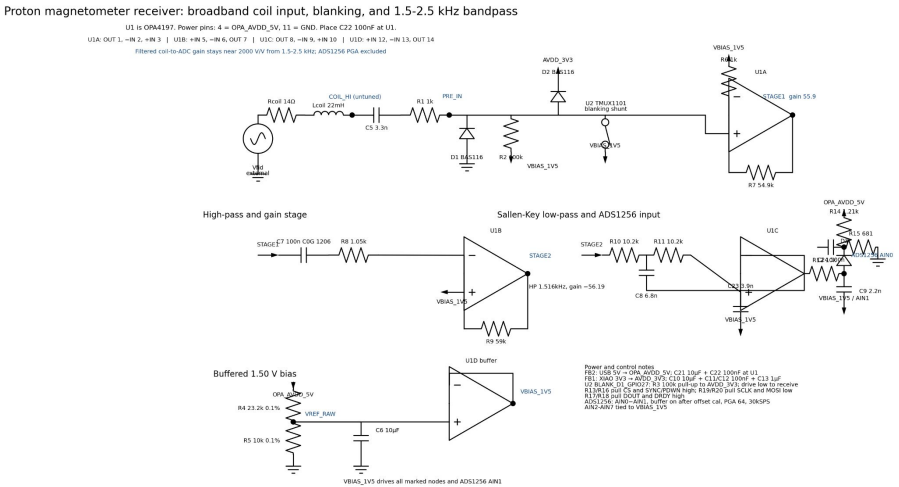
Design

- Untuned coil, so 1.7 kHz and 2.1 kHz both pass
- About 2000 V/V from coil to the ADC pins, before the PGA
- Passband 1.5–2.5 kHz
- ADC input is (AIN0-AIN1) with PGA 64 at 30 kSPS



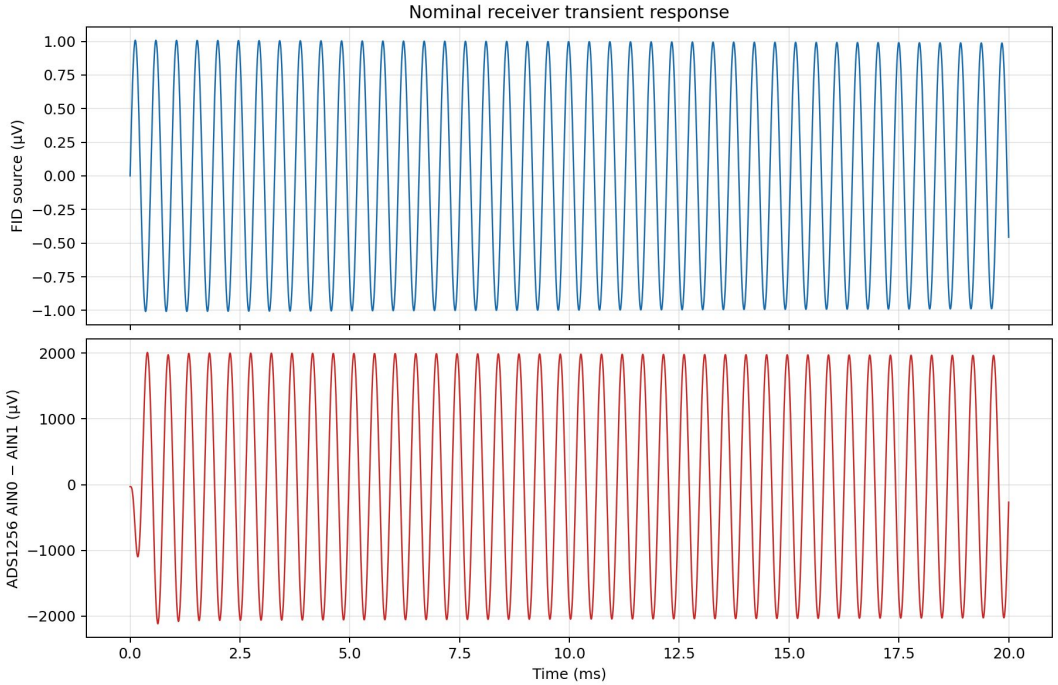
Analog stages

- High-pass before gain, so 50/60 Hz is attenuated first
- U1A and U1B set 55.9 × 56.2 = 3141 V/V
- Filters bring the 1.5–2.5 kHz band down to about 2000 V/V
- Sallen–Key natural frequency is 3.03 kHz
- No notch at 1.8 kHz; that is a valid proton signal



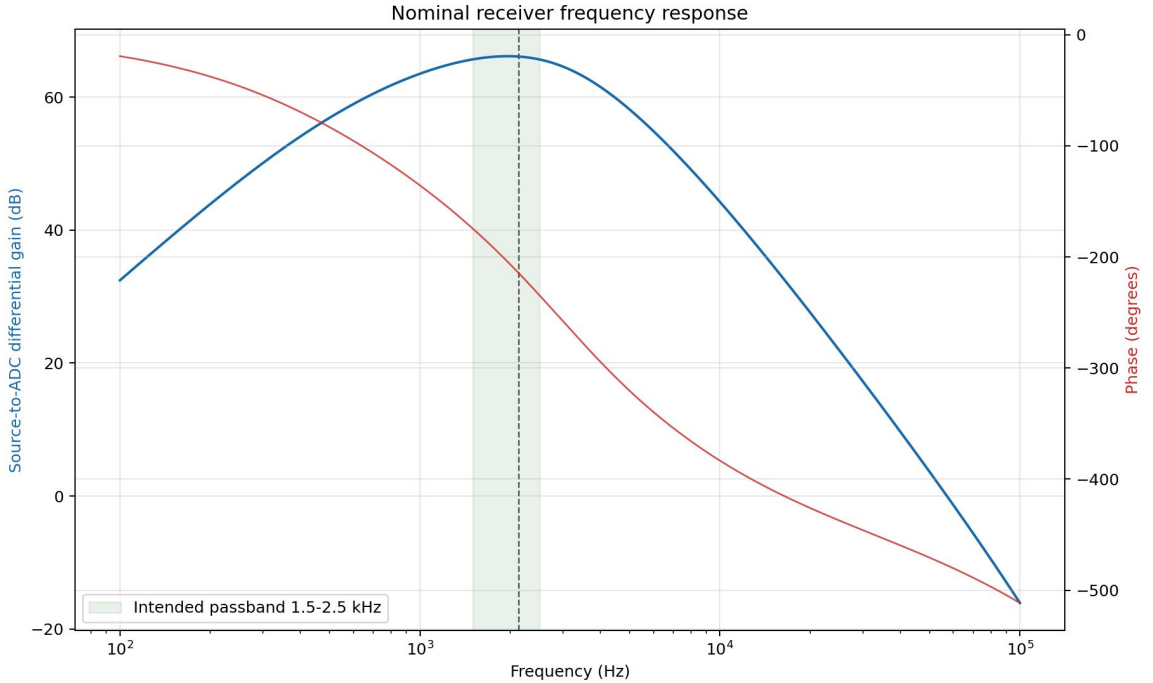
Transient response

- Simulated, not measured
- About 1 μV peak at the coil, 2128.8 Hz
- Differential ADC input settles near ±2.0 mV
- PGA 64 full scale is ±78 mV



Frequency response

- Simulated, not measured
- In-band gain 1924-2032 V/V
- 2020 V/V at 2128.8 Hz
- 7.3 V/V at 30 kHz, 49 dB below the passband



Frequency estimator (MCU code written with AI tools)

- One record, up to 1.5 s at 30 kSPS
- Full-rate samples are not stored; only the decimated envelope is kept
- Seed: Goertzel over 500–3500 Hz, then a 3-bin parabolic refine
- Mix with a digital oscillator at that seed
- Low-pass and decimate by 10
- Fit the envelope decay, and weight the tail down
- Scan residual frequency at 0.02 Hz steps
- Parabolic refine of the zoom peak
- A peak on either end of the seed band or the ±20 Hz grid is rejected

Next Steps

- Order components
- Build the circuit on a breadboard and test it